



End-to-End Data Analytics Projects

Instructions :

Choose **any ONE** of the following projects.

Your project should include the complete data analytics workflow:

- Data Collection using **Requests + BeautifulSoup**
- Use **Headers** wherever required
- Scrape data from **multiple pages** (if available)
- Change URLs dynamically wherever required
- Data Cleaning using **Pandas** and **Regular Expressions**
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Business Insights
- Business Recommendations

Project 1 - Bike Market Analysis

Website:

<https://www.bikewale.com/>

Example:

<https://www.bikewale.com/tvs-bikes/>

Objective

Scrape **all available bike brands** from the website.

For each brand, dynamically change the URL and scrape the bike details.

Extract

- Bike Name
- Brand
- Price
- Mileage
- Engine Capacity
- Transmission
- Kerb Weight
- Fuel Tank Capacity
- User Rating

Data Cleaning

- Remove currency symbols
- Convert price into numeric
- Extract engine capacity (cc)
- Extract mileage (kmpl)
- Extract fuel tank capacity (litres)
- Convert ratings into float

Feature Engineering

Create

- Price Category
- Engine Category
- Mileage Category

Perform Analysis

- Brand-wise average price
- Brand-wise average mileage
- Highest rated bikes
- Mileage vs Price
- Engine Capacity vs Price
- Most affordable brands
- Premium bike brands
- Price distribution
- Engine capacity distribution
- Top 10 expensive bikes

Project 2 - New Car Market Analysis

Website:

<https://www.carwale.com/>

Example:

<https://www.carwale.com/tata-cars/>

Objective

Scrape **all available car brands**.

Change the URL dynamically for every brand and collect the data.

Extract

- Car Name
- Brand
- Price
- Fuel Type
- Transmission
- Engine
- Mileage
- Seating Capacity
- User Rating

Data Cleaning

- Remove currency symbols
- Convert prices into numeric
- Extract engine capacity
- Extract mileage
- Convert ratings into float

Analysis

- Brand-wise average price
- Fuel type comparison
- Manual vs Automatic

- Mileage comparison
 - Engine comparison
 - Most expensive brands
 - Cheapest brands
 - Rating analysis
 - Premium vs Budget cars
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Project 3 - Used Car Market Analysis

Website:

<https://www.cars24.com/>

Example:

<https://www.cars24.com/buy-used-toyota-cars/>

Objective

Scrape used cars by changing

- Brand
- City

Example

Toyota → Hyundai → Tata → Honda

Hyderabad → Bangalore → Chennai → Delhi

Automatically generate the URLs and scrape the data.

Extract

- Car Name
- Brand
- City
- Model Year
- Kilometers Driven
- Fuel Type

- Transmission
- Price
- Owner Type

Data Cleaning

- Remove currency symbols
- Convert kilometres into integers
- Extract manufacturing year
- Convert price into numeric

Analysis

- Average price by city
 - Average price by brand
 - Fuel type comparison
 - Manual vs Automatic
 - Owner type distribution
 - Kilometer driven distribution
 - Brand popularity
 - Used car price comparison
 - City-wise price comparison
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Project 4 - Smartphone Market Analysis

Website

Flipkart / Amazon

Extract

- Brand
- Price
- Discount
- Rating
- RAM
- Storage
- Battery
- Camera

Analysis

- Brand-wise pricing
 - RAM vs Price
 - Storage comparison
 - Battery comparison
 - Discount analysis
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Project 5 - Laptop Market Analysis

Website

Flipkart / Amazon

Extract

- Brand
- Processor
- RAM
- SSD
- Display
- Price
- Discount
- Rating

Analysis

- Processor comparison
 - RAM comparison
 - SSD comparison
 - Brand comparison
 - Premium laptops
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Project 6 - Washing Machine Market Analysis

Website

Flipkart

Extract

- Brand
- Capacity
- Price
- Warranty
- Power Rating
- RPM
- Discount
- Rating
- Top Load / Front Load

Analysis

- Capacity vs Price
 - Warranty analysis
 - Brand comparison
 - Rating comparison
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Project 7 - Television Market Analysis

Website

Flipkart

Extract

- Brand
- Screen Size
- Resolution
- Smart TV
- Price
- Discount
- Rating

Analysis

- Resolution analysis
 - Screen size comparison
 - Brand comparison
 - Smart TV comparison
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Project 8 - Smartwatch Market Analysis

Website

Flipkart / Amazon

Extract

- Brand
- Price
- Display
- Battery
- Rating
- Discount

Analysis

- Brand comparison
 - Price comparison
 - Battery comparison
 - Rating comparison
-

Project 9 - Books Market Analysis

Website

<http://books.toscrape.com>

Analysis

- Category-wise pricing
 - Rating analysis
 - Availability analysis
 - Price distribution
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Project 10 - IMDb Top Movies Analysis

Extract

- Movie
- Rating
- Year
- Runtime

Analysis

- Movies by decade
 - Rating comparison
 - Highest rated movies
 - Runtime analysis
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Common Requirements (Mandatory)

1. Web Scrapping

- Use **Requests** and **BeautifulSoup**
- Use **HTTP Headers** wherever required
- Scrape data from multiple pages (where applicable)
- Dynamically change URLs (brand/page/city/category) wherever applicable

2. Data Cleaning

- Handle missing values

- Remove duplicates
- Use **Regular Expressions** for at least **5 data-cleaning or extraction tasks**
- Convert data into appropriate data types

3. Feature Engineering

Create at least **3 new columns** from the scraped data.

4. Exploratory Data Analysis

Include:

- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis
- At least **8 visualizations**

5. Business Questions

Answer **10 business questions** using your analysis.

6. Business Insights

Write **8 meaningful insights**.

7. Recommendations

Provide **5 actionable recommendations** based on your findings.

Submissions:

- Jupyter Notebook ([.ipynb](#))
- Scraped Dataset ([.csv](#))
- Cleaned Dataset ([.csv](#))
- Visualizations
- Business Questions with Answers
- Business Insights
- Recommendations

UPLOAD ALL THESE IN A ZIPPED FOLDER ON THE LMS



INNOMATICS
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